Wenyu Huang

LLM / NLP Researcher

Edinburgh, UK | wyhuang1998@gmail.com | w.huang@ed.ac.uk | +44 7759 673027

hwy9855.github.io | GitHub | Google Scholar

RESEARCH SUMMARY

Final-year PhD researcher in Natural Language Processing at the University of Edinburgh, studying how language models retrieve, represent, and reason with external knowledge across retrieval-augmented generation, memory, multi-hop question answering, and language agents. First-author publications at ACL, EMNLP, SIGIR, IJCNLP-AACL, and Knowledge-Based Systems, including an ACL 2025 oral paper, with industrial research experience at Microsoft Research Cambridge and Huawei R&D UK.

EDUCATION

Ph.D. in Natural Language Processing — **2021–present**
University of Edinburgh, UKRI Centre for Doctoral Training in NLP — *Edinburgh, UK*
Thesis submission expected Aug 2026; viva expected late 2026/early 2027.
Supervisors: Prof. Jeff Z. Pan and Prof. Mirella Lapata.

M.Sc. in Artificial Intelligence — **2020–2021**
University of Edinburgh — *Edinburgh, UK*
Graduated with Distinction.

B.Eng. in Software Engineering — **2016–2020**
South China University of Technology — *Guangzhou, China*

RESEARCH EXPERIENCE

Doctoral Researcher — **2021–present**
University of Edinburgh, Institute for Language, Cognition and Computation — *Edinburgh, UK*

- Developed a 220M-parameter generative subgraph retriever competitive with 7B-parameter baselines under the published settings; a 3B retriever paired with an LLM reader reported state-of-the-art results at publication on WebQSP and ComplexWebQuestions. [paper] [code]
- Analysed order sensitivity across causal and bidirectional attention settings in multi-hop QA, and developed bidirectional-attention variants that improved reasoning over permuted evidence. [paper] [code]
- Created LTGen, a benchmark for question answering over long-tail facts, and evaluated how knowledge-graph evidence affects factual coverage and hallucination. [paper]

Research Intern — **Apr–Jun 2026**
Microsoft Research Cambridge, Machine Intelligence — *Cambridge, UK*

- Investigated whether external factual memory can improve the parameter efficiency of language-model pretraining, working with John Winn in the Machine Intelligence group.
- Ran multi-node LLM pretraining and fine-tuning experiments with PyTorch FSDP2, scaling to 32 NVIDIA B200 GPUs across four nodes.

Research Intern — **Jul 2025–Mar 2026**
Huawei Technologies R&D (UK) Ltd — *Edinburgh, UK*

- Led research on reinforcement-learning post-training for search agents using veRL, resulting in the first-author manuscript *Beyond Outcome Rewards: A Study of Intermediate Supervision for Search Agents*, currently under review.

Research Intern — **Jun–Nov 2022**
Huawei Technologies R&D (UK) Ltd — *Edinburgh, UK*

- Researched algorithms for large-scale entity alignment.

TECHNICAL SKILLS

LLM Training and Post-training: Multi-node LLM pretraining and fine-tuning at scales of up to 32 NVIDIA B200 GPUs across four nodes; PyTorch FSDP2, veRL, and parameter-efficient fine-tuning (PEFT/QLoRA).

LLM Inference and Systems: vLLM, SGLang, and Hugging Face Transformers.

Retrieval, Agents, and Memory: Retrieval-augmented generation, structured retrieval, knowledge graphs, agentic reinforcement learning, memory-augmented language models, long-context reasoning, and stateful tool use.

Research and Evaluation: Benchmark design, controlled ablations, error analysis, retrieval-reader decomposition, context interventions, and reproducible offline evaluation.

Programming and Developer Tools: Python, Git, Linux; agent-assisted implementation, debugging, testing, and code review with Claude Code, OpenAI Codex, and GitHub Copilot.

SELECTED PUBLICATIONS

[1] **Masking in Multi-hop QA: An Analysis of How Language Models Perform with Context Permutation**
Wenyu Huang, Pavlos Vougiouklis, Mirella Lapata, and Jeff Z. Pan. *ACL 2025 (Main Conference), Oral*. [code]

[2] **Less is More: Making Smaller Language Models Competent Subgraph Retrievers for Multi-hop KGQA**
Wenyu Huang, Guancheng Zhou, Hongru Wang, Pavlos Vougiouklis, Mirella Lapata, and Jeff Z. Pan. *Findings of EMNLP 2024*. [code]

[3] **Prompting Large Language Models with Knowledge Graphs for Question Answering Involving Long-tail Facts**
Wenyu Huang, Guancheng Zhou, Mirella Lapata, Pavlos Vougiouklis, Sebastien Montella, and Jeff Z. Pan. *Knowledge-Based Systems, 2025*.

[4] **Rethinking Memory in AI: Taxonomy, Operations, Topics, and Future Directions**
Yiming Du*, **Wenyu Huang***, Danna Zheng*, et al. *arXiv, 2025*. *Equal contribution.

[5] **A Large-scale Offer Alignment Model for Partitioning, Filtering and Matching Product Offers**
Wenyu Huang, André Melo, and Jeff Z. Pan. *SIGIR 2024*.

[6] **Retrieval Augmented Generation with Rich Answer Encoding**
Wenyu Huang, Mirella Lapata, Pavlos Vougiouklis, Nikos Papasarantopoulos, and Jeff Z. Pan. *IJCNLP-AACL 2023*. [code]

ADDITIONAL PUBLICATIONS

[7] Yiming Du, Baojun Wang, Yifan Xiang, Zhaowei Wang, **Wenyu Huang**, et al. Memory-T1: Reinforcement Learning for Temporal Reasoning in Multi-session Agents. *ICLR 2026, Poster*.

[8] Hongru Wang, **Wenyu Huang**, Yufei Wang, et al. Rethinking Stateful Tool Use in Multi-Turn Dialogues: Benchmarks and Challenges. *Findings of ACL 2025*.

[9] Hongru Wang, **Wenyu Huang**, Yang Deng, et al. UniMS-RAG: A Unified Multi-source Retrieval-Augmented Generation for Personalized Dialogue Systems. *arXiv, 2024*.